

CHAPTER 3

MOTION

1.2 Motion

- 1 Define speed as distance travelled per unit time and define velocity as change in displacement per unit time
- 2 Recall and use the equation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$v = \frac{s}{t}$$

- 3 Recall and use the equation

$$\text{average speed} = \frac{\text{total distance travelled}}{\text{total time taken}}$$

- 4 Define acceleration as change in velocity per unit time; recall and use the equation

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

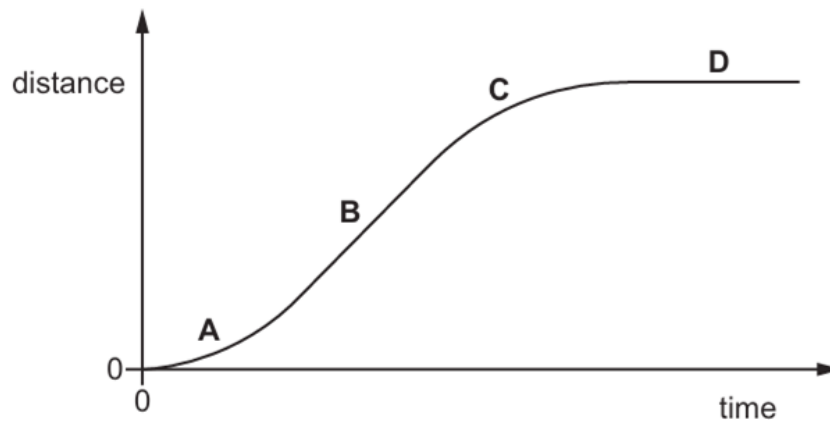
$$a = \frac{\Delta v}{\Delta t}$$

- 5 State what is meant by, and describe examples of, uniform acceleration and non-uniform acceleration
- 6 Know that a deceleration is a negative acceleration and use this in calculations
- 7 Sketch, plot and interpret distance–time and speed–time graphs
- 8 Determine from the shape of a distance–time graph when an object is:
 - (a) at rest
 - (b) moving with constant speed
 - (c) accelerating
 - (d) decelerating
- 9 Determine from the shape of a speed–time graph when an object is:
 - (a) at rest
 - (b) moving with constant speed
 - (c) moving with constant acceleration
 - (d) moving with changing acceleration
- 10 State that the acceleration of free fall g for an object near to the surface of the Earth is approximately constant and is approximately 9.8 m/s^2
- 11 Calculate speed from the gradient of a distance–time graph
- 12 Calculate the area under a speed–time graph to determine the distance travelled for motion with constant speed or constant acceleration
- 13 Calculate acceleration from the gradient of a speed–time graph

1. O/N/23/11/Q2

The diagram shows a distance–time graph for a car travelling in a straight line.

In which region is the car decelerating?



2. O/N/23/11/Q4

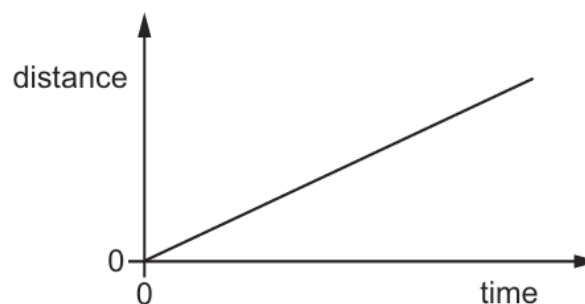
A car begins to move. It speeds up until it reaches a constant speed.

What happens to the acceleration and what happens to the velocity of the car?

- A** Both the acceleration and the velocity change.
- B** Only the acceleration changes.
- C** Only the velocity changes.
- D** Neither the acceleration nor the velocity changes.

3. O/N/23/12/Q2

The diagram shows the distance–time graph of a car.



The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

- A** acceleration
- B** displacement
- C** gravitational potential energy
- D** kinetic energy

4. O/N/23/12/Q3

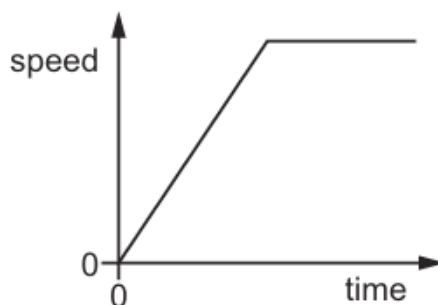
A distance–time graph and a speed–time graph show the motion of an object.

How are the speed and the acceleration of the object determined from these graphs?

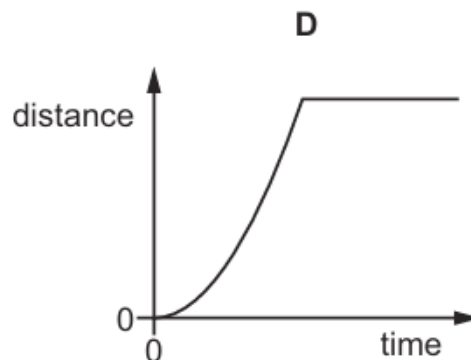
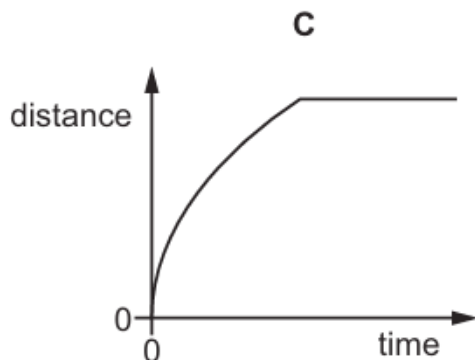
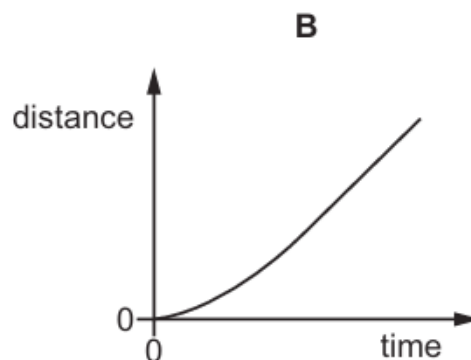
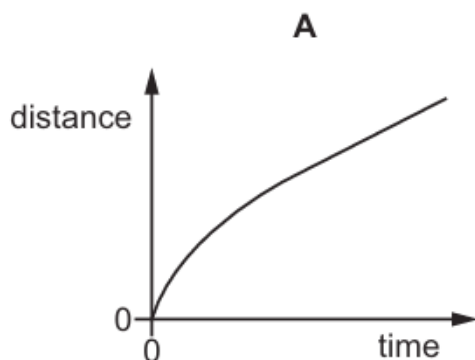
	speed	acceleration
A	area under the distance–time graph	area under the speed–time graph
B	area under the distance–time graph	gradient of the speed–time graph
C	gradient of the distance–time graph	area under the speed–time graph
D	gradient of the distance–time graph	gradient of the speed–time graph

5. M/J/23/11/Q4

A speed–time graph for the journey of a car is shown.

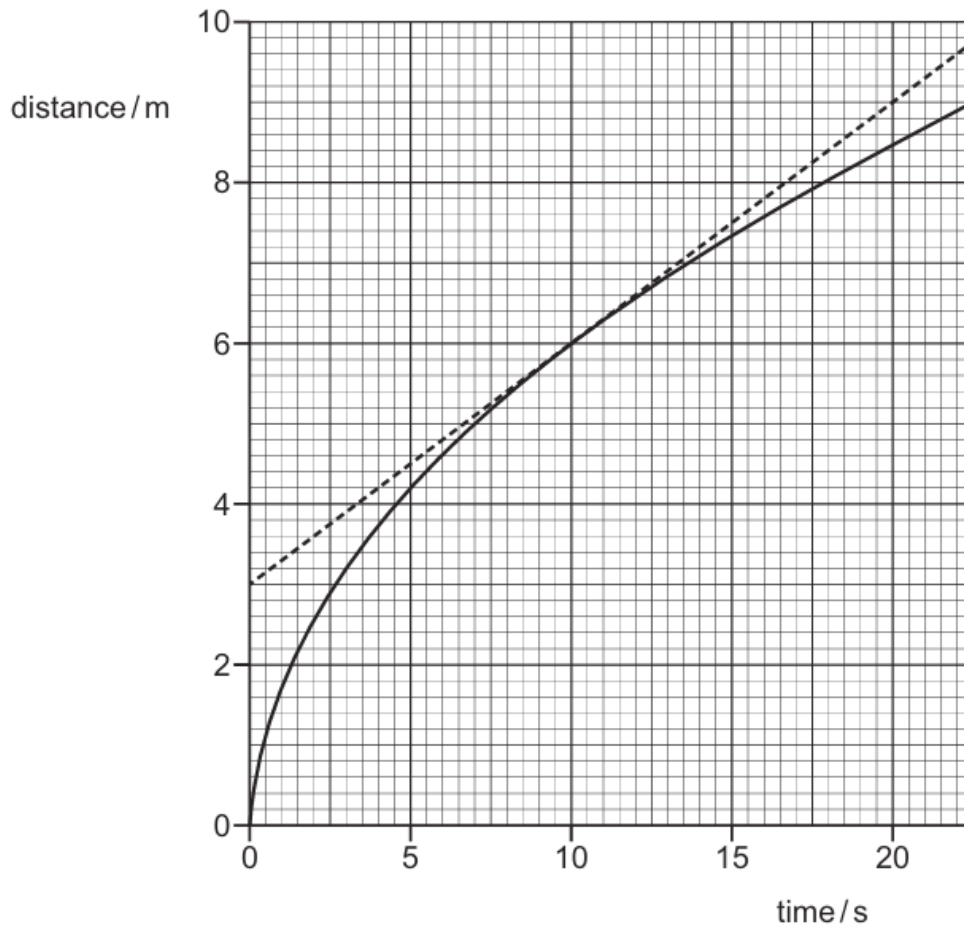


What is the distance–time graph for the journey?



6. M/J/23/12/Q4

The curved line on the distance–time graph shows the motion of a toy car. The straight line is the tangent to the curve at 10 s.



What is the speed of the toy car at 10 s?

- A** 0.3 m/s **B** 0.6 m/s **C** 1.7 m/s **D** 3.3 m/s

7. O/N/22/11/Q3

A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.

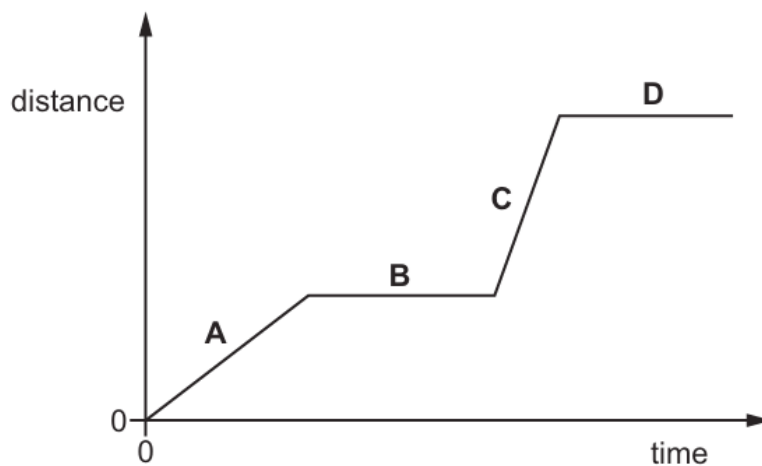
The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.

How far does the car travel in the sand before coming to rest?

- A** 2.0 m **B** 4.0 m **C** 20 m **D** 40 m

8. O/N/22/11/Q4

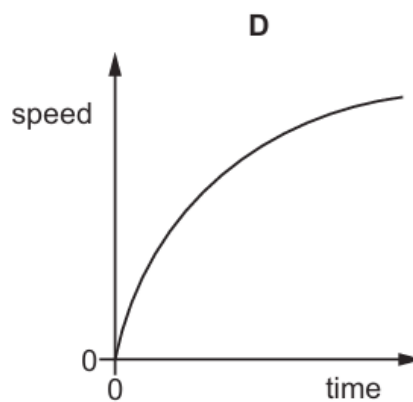
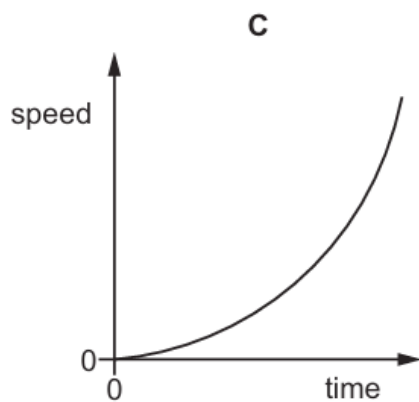
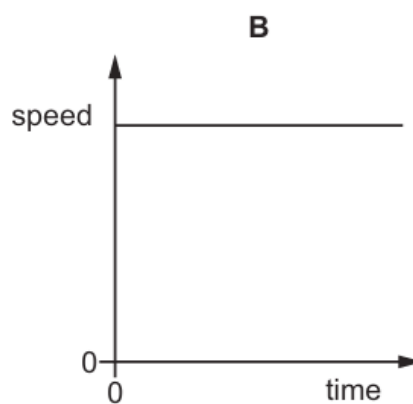
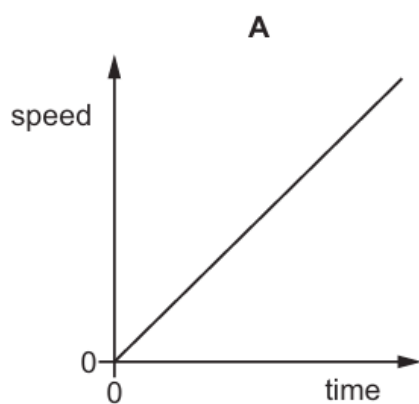
Which part of the distance–time graph shows the fastest constant speed?



9. O/N/22/11/Q5

An object is accelerating along a straight, horizontal track. The acceleration is uniform.

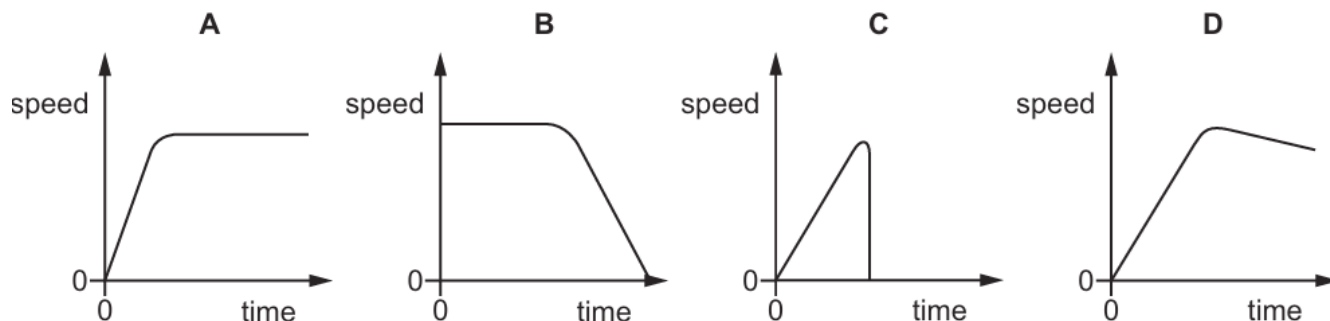
Which speed–time graph represents the motion of the object?



10. O/N/22/12/Q5

A ball starts from rest and rolls down a steep slope. The ball then rolls along rough horizontal ground.

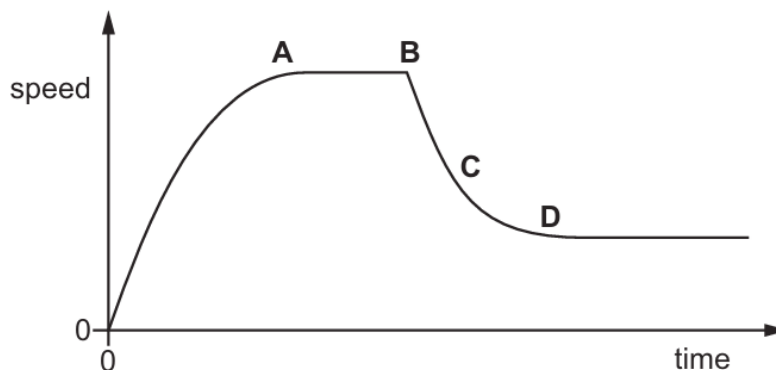
Which graph shows the speed of the ball at different times?



11. M/J/22/12/Q3

The graph shows the speed-time graph for a parachutist who jumps from a plane but does not open his parachute immediately.

At which point does he open his parachute?



12. O/N/21/11/Q3

Two identical objects begin to fall from rest.

One object falls from 200 m above the Earth's surface and the other falls from 200 m above the Moon's surface. One second after they have started to fall, both objects are still accelerating.

There is no atmosphere on the Moon and the weight of each object is constant.

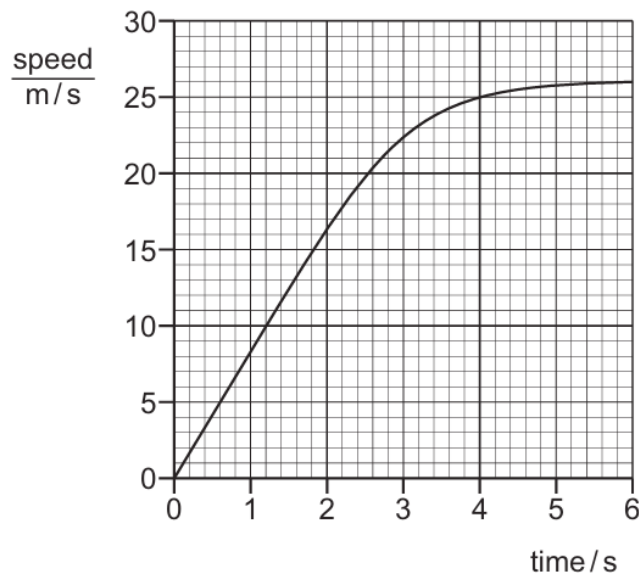
Which row describes the motion of both objects at this time?

	acceleration of object falling above the Earth's surface	acceleration of object falling above the Moon's surface
A	decreasing	constant
B	constant	constant
C	increasing	decreasing
D	increasing	increasing

13. O/N/21/12/Q3

A skier slides down a slope.

The diagram shows how his speed varies with time.

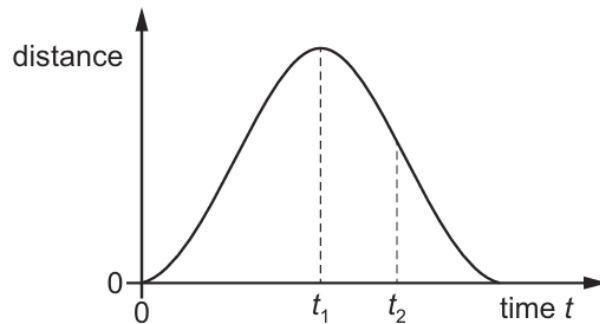


What is his average acceleration during the 6.0 s?

- A** 2.2 m/s^2 **B** 4.3 m/s^2 **C** 8.0 m/s^2 **D** 13.0 m/s^2

14. M/J/21/11/Q3

A train sets off from a station at time $t = 0$. The graph shows how the distance between the train and the station varies with time.



Which statement about the movement of the train between time t_1 and t_2 is correct?

- A** Its speed is decreasing and it is moving away from the station.
- B** Its speed is decreasing and it is moving towards the station.
- C** Its speed is increasing and it is moving away from the station.
- D** Its speed is increasing and it is moving towards the station.

15. O/N/20/11/Q2

A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

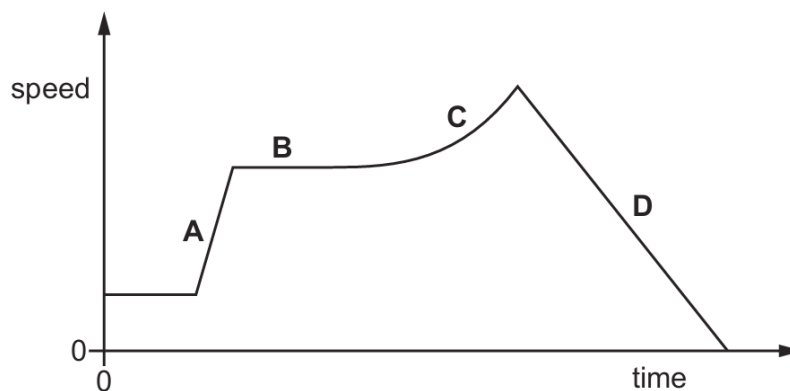
What happens to the acceleration and what happens to the velocity of the car during the journey?

- A** Both the acceleration and the velocity change.
- B** Only the acceleration changes.
- C** Only the velocity changes.
- D** Neither the acceleration nor the velocity changes.

16. O/N/20/12/Q3

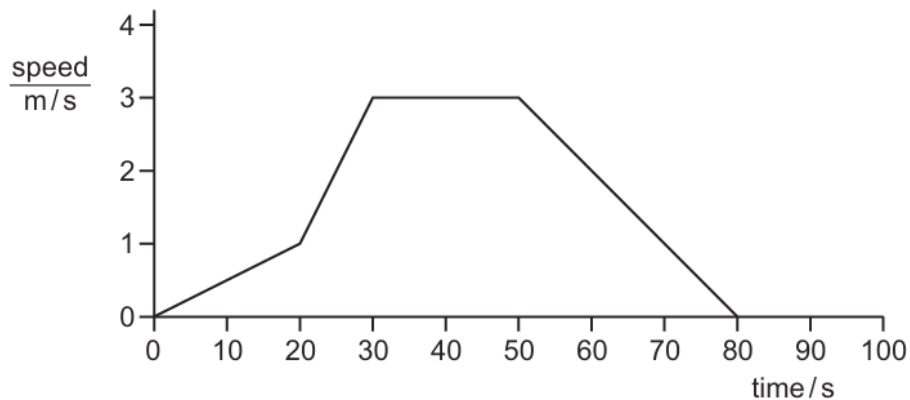
The speed–time graph for a car's journey is shown.

During which part of the journey is the car moving with non-uniform acceleration?



17. M/J/20/11/Q3

The graph shows a short journey. It is a speed–time graph.



What is the greatest speed reached?

- A** 1 m/s **B** 2 m/s **C** 3 m/s **D** 4 m/s

18. M/J/20/12/Q5

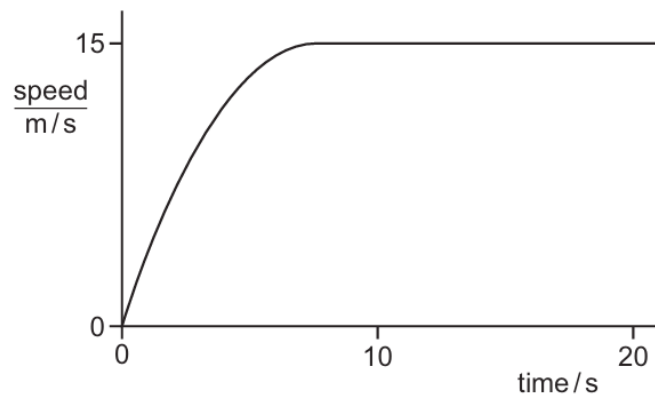
Two cameras are a known distance apart. The exact time that a vehicle passes each of the cameras is recorded.

What can be obtained from the information?

- A** average acceleration of the vehicle
B average speed of the vehicle
C maximum acceleration of the vehicle
D maximum speed of the vehicle

19. O/N/19/11/Q5

The graph shows how the speed of a car varies with time.



Which statement about the acceleration of the car between the times 10 s and 20 s is correct?

- A** The acceleration decreases.
B The acceleration increases.
C The acceleration is constant, but not zero.
D The acceleration is zero.

20. O/N/19/12/Q5

Part of a speed–time graph for two runners is shown. They are running along the same track.



Which statement **must** be correct?

- A Both runners started at the same moment.
- B Runner 1 has a greater acceleration than runner 2.
- C Runner 1 is moving faster than runner 2.
- D The distance between the two runners stays constant.

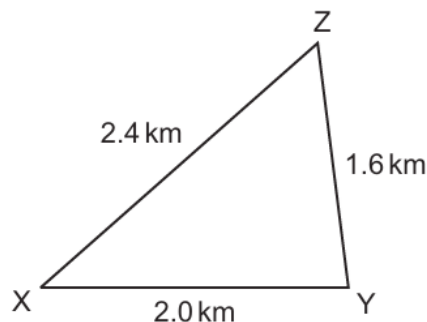
21. O/N/19/12/Q7

In which descent is the acceleration equal to the acceleration of free fall g at all times?

- A a bungee jumper leaping from a bridge
- B a feather falling in a vertical tube that contains a vacuum
- C a hailstone travelling to Earth at terminal velocity
- D a sky-diver dropping from an aircraft towards the ground

22. M/J/19/12/Q2

A woman runs 2.0 km from X to Y in 20 minutes and then rests at Y for 10 minutes. She then takes 10 minutes to run 1.6 km from Y to Z.



What is the size of her average velocity for the journey from X to Z?

- A 0.060 km/minute
- B 0.080 km/minute
- C 0.090 km/minute
- D 0.130 km/minute

23. M/J/19/12/Q3

A cyclist travelling in a straight line at 8.0 m/s accelerates to 12 m/s in a time of 6.0 s.

Which expression gives the cyclist's acceleration?

A $\frac{6}{12-8}$

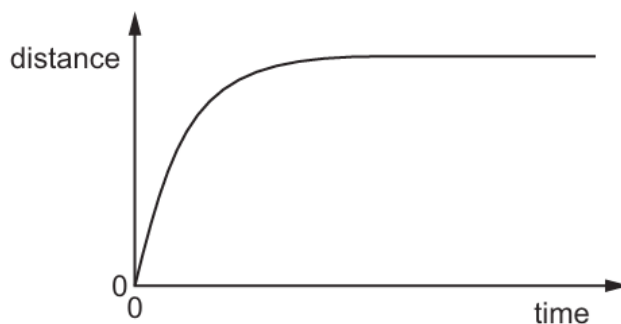
B $\frac{8}{12-6}$

C $\frac{12-6}{8}$

D $\frac{12-8}{6}$

24. M/J/19/12/Q4

The diagram shows the distance–time graph for a moving object.



What is the moving object?

- A** a ball thrown vertically upwards and falling back to the thrower
- B** a car, starting from rest, speeding up as it moves away from a traffic signal
- C** a rock dropped from a high cliff, falling into the sea below
- D** a train braking to a halt as it stops at a station

25. O/N/18/12/Q2

The table shows how the speeds of four bodies, **A**, **B**, **C** and **D**, change with time.

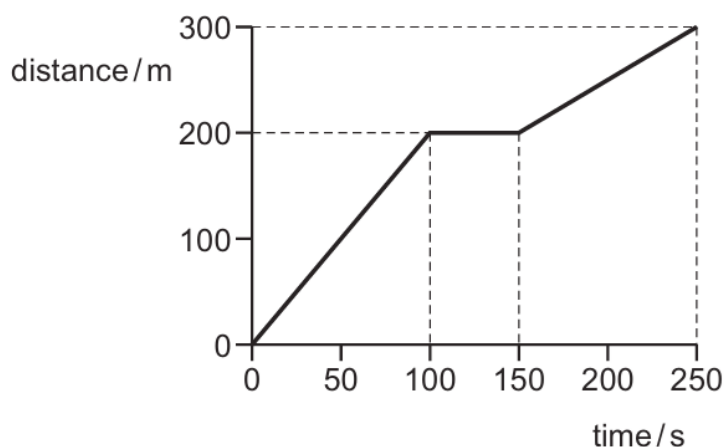
Which body has an acceleration that is **not** constant?

time/s	speed of A m/s	speed of B m/s	speed of C m/s	speed of D m/s
0	0	0	0	5.5
1	1.0	2.0	3.0	6.5
2	3.0	4.0	6.0	7.5
3	6.0	6.0	9.0	8.5

26. O/N/18/11/Q4

A cyclist takes a ride lasting 250 s.

The graph shows how the distance from the starting position varies with time.



What is his average speed for the whole journey?

- A** 1.0 m/s **B** 1.2 m/s **C** 1.5 m/s **D** 2.0 m/s

27. M/J/18/12/Q11

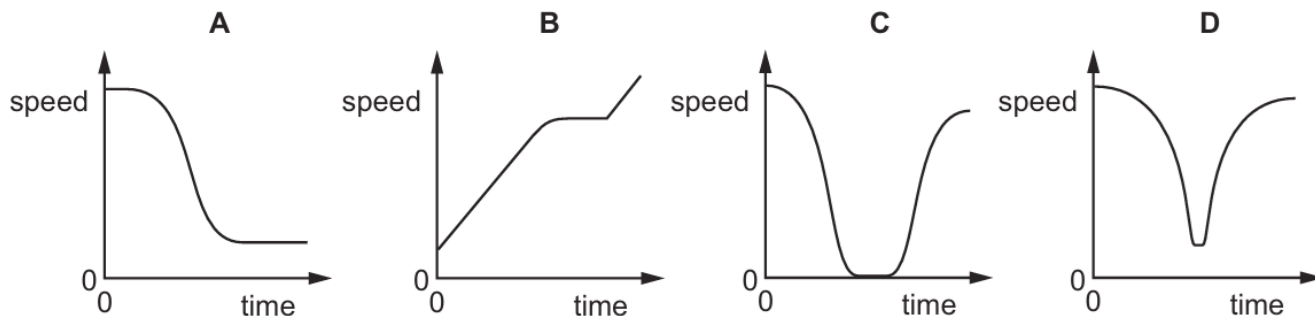
A force of 1600 N accelerates a car, of mass 800 kg, from rest.

What is the car's acceleration and its velocity after 4.0 s?

	<u>acceleration</u> m/s ²	<u>velocity</u> m/s
A	0.50	0.13
B	0.50	0.50
C	2.0	2.0
D	2.0	8.0

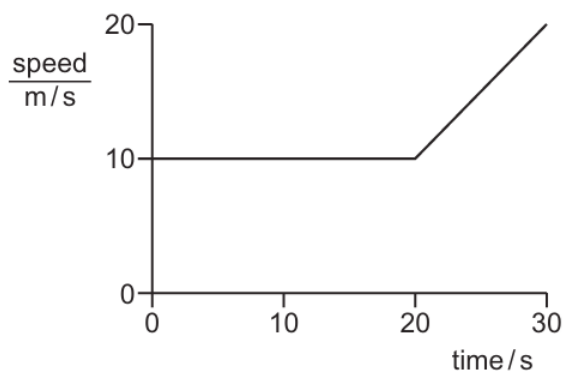
28. M/J/18/11/Q4

Which speed-time graph represents the motion of a railway train making a short stop at a station?



29. O/N/17/11/O4

An object travels for 20s with a constant speed of 10m/s. For the next 10s, it accelerates uniformly to 20m/s.

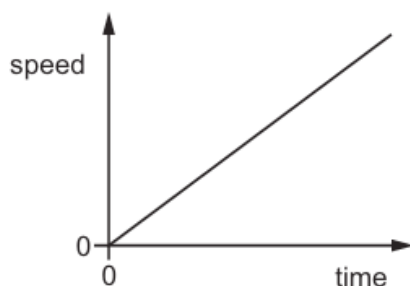


What is the total distance travelled by the object in the 30 s?

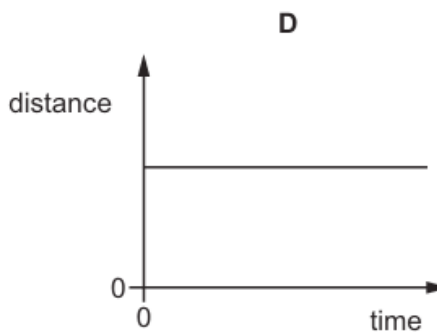
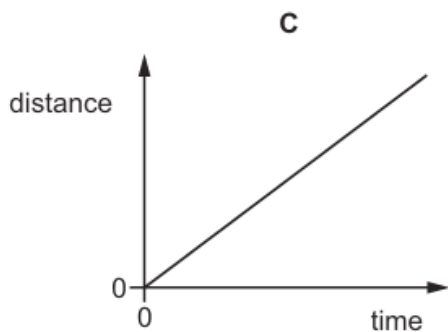
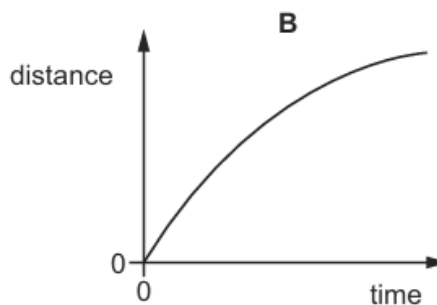
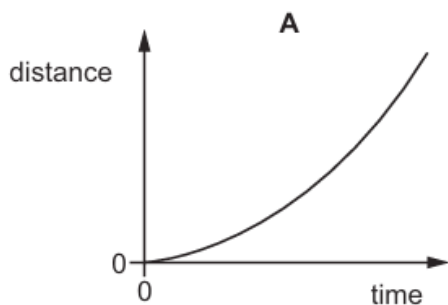
- A** 300 m **B** 350 m **C** 400 m **D** 450 m

30. O/N/17/12/Q3

The speed-time graph represents a short journey.

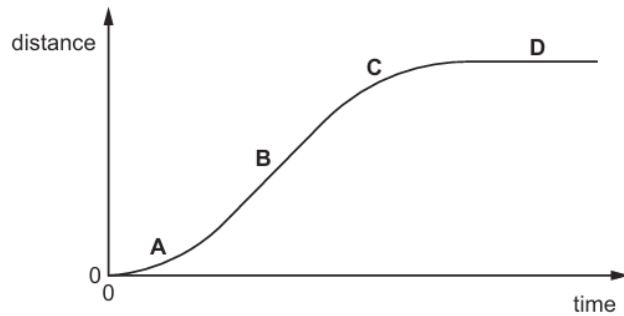


Which distance-time graph represents the same journey?



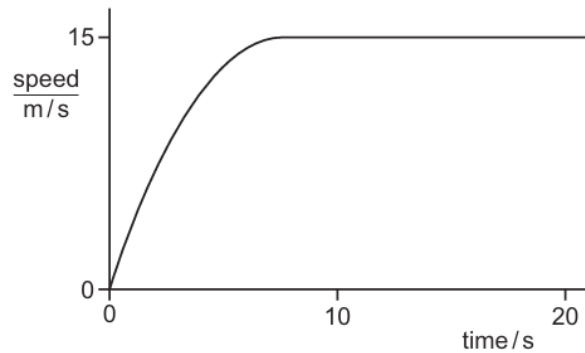
31. O/N/16/12/Q4

The graph shows a distance-time graph for a car travelling in a straight line.
In which region is the car decelerating?



32. M/J/16/11/Q4

The graph shows how the speed of a car varies with time.



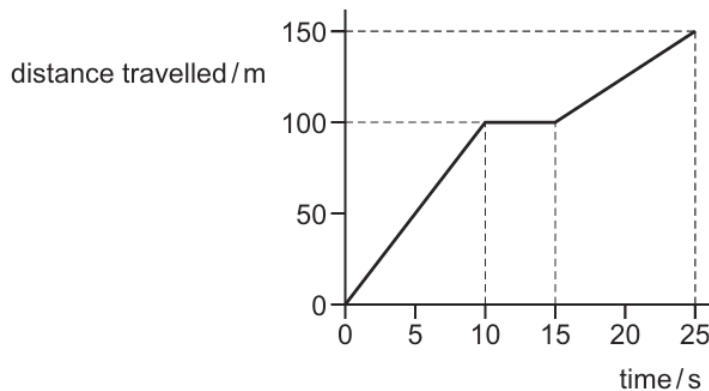
Which statement about the acceleration of the car between 10 s and 20 s is correct?

- | | |
|--------------------------------------|--|
| A The acceleration decreases. | C The acceleration is constant, but not zero. |
| B The acceleration increases. | D The acceleration is zero. |

33. M/J/15/12/Q3

A cyclist takes a ride lasting 25 s.

The diagram shows how her distance travelled from the starting position varies with time.

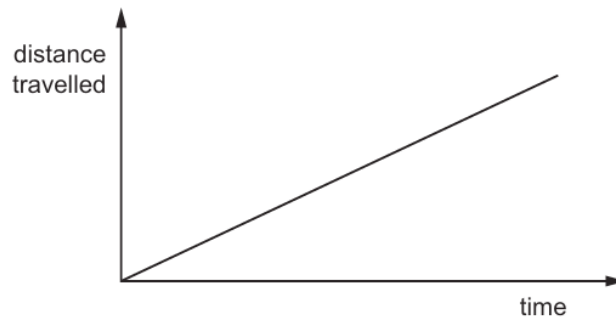


What is her average speed for the whole ride?

- | | | | |
|------------------|------------------|-------------------|-------------------|
| A 6.0 m/s | B 7.5 m/s | C 10.0 m/s | D 11.0 m/s |
|------------------|------------------|-------------------|-------------------|

34. M/J/14/12/Q6

The distance travelled by a car is increasing uniformly as it is driven along a straight road up a hill.



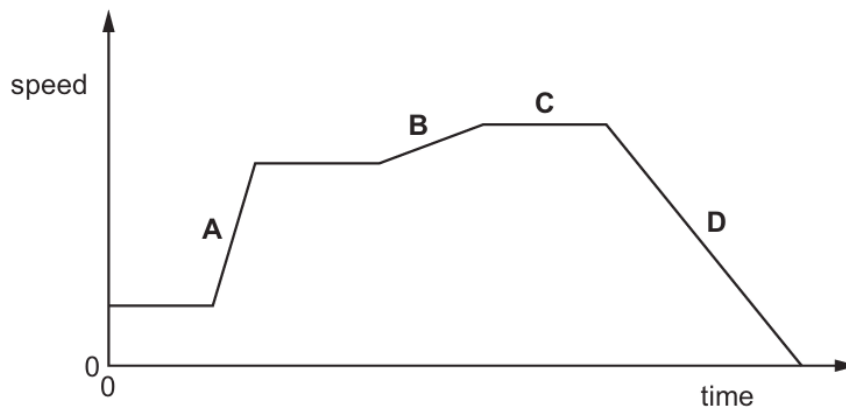
Which quantity for the car is constant but not zero?

- | | |
|---|--------------------------|
| A acceleration | C kinetic energy |
| B gravitational potential energy | D resultant force |

35. M/J/14/11/Q3

The graph shows how the speed of a car travelling in a straight line changes with time.

Which section shows the largest acceleration?



36. O/N/13/12/Q2

An object moves from P to Q in 10 s with uniform acceleration.

velocity at P = 5 m/s

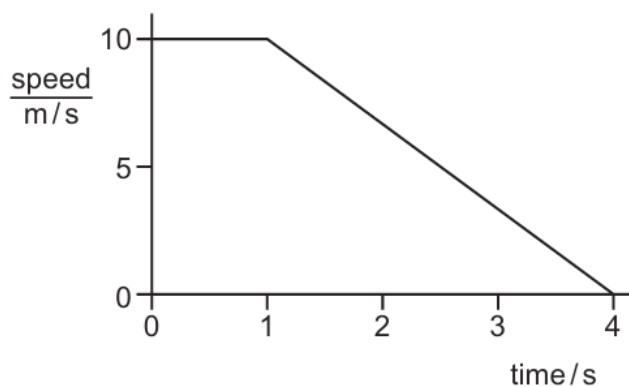
velocity at Q = 12 m/s

What is the acceleration?

- | | | | |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| A 0.5 m/s ² | B 0.7 m/s ² | C 1.2 m/s ² | D 1.7 m/s ² |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|

37. M/J/13/12/Q4

The diagram shows the speed-time graph of the motion of a car for four seconds.

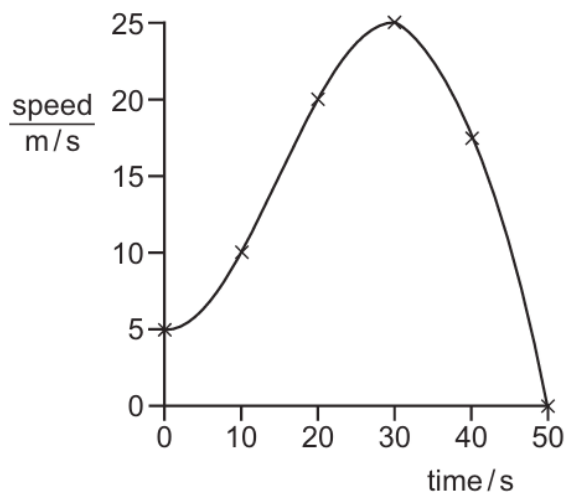


What is the distance travelled by the car in the four seconds?

- A** 15 m **B** 25 m **C** 30 m **D** 40 m

38. O/N/12/12/Q4

The speed-time graph for a car is shown.

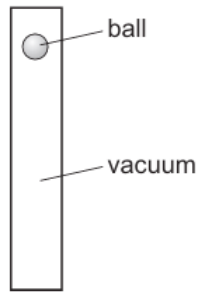


What is the acceleration of the car at 30 s?

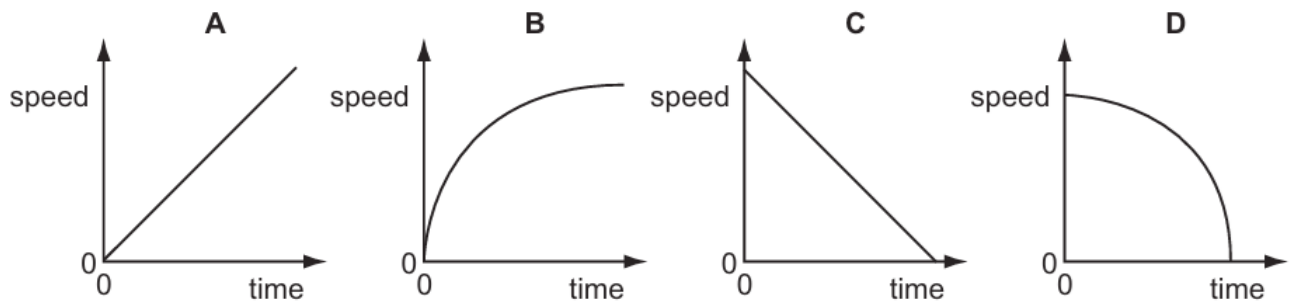
- A** 0 **B** $\frac{25-5}{30} \text{ m/s}^2$ **C** $\frac{25}{30} \text{ m/s}^2$ **D** $\frac{25}{50} \text{ m/s}^2$

39. O/N/12/12/Q3

A table-tennis ball is released from the top of an evacuated tube.



Which graph shows how the speed of the ball changes with time as it falls to the bottom of the tube?



40. O/N/12/11/Q4

A car is stationary at a set of traffic lights.

When the lights turn green, the car begins to move and continues to speed up until it reaches the maximum speed allowed. It continues to travel at this constant speed for the rest of the journey.

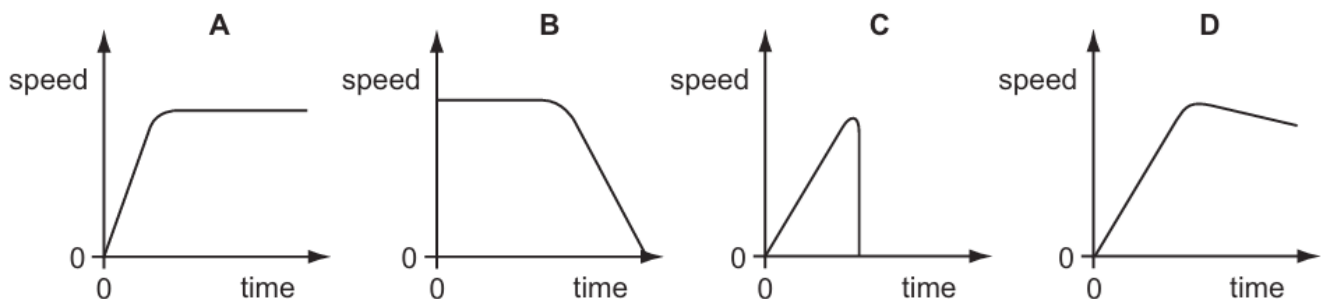
What happens to the acceleration and to the velocity of the car during this journey?

- A** Both the acceleration and the velocity change.
- B** Only the acceleration changes.
- C** Only the velocity changes.
- D** Neither the acceleration nor the velocity changes.

41. O/N/12/11/Q3

A ball starts to roll down a steep slope and then along rough horizontal ground.

Which graph best shows the **speed** of the ball?



42. M/J/12/12/Q4

A student drops a table-tennis ball in air.

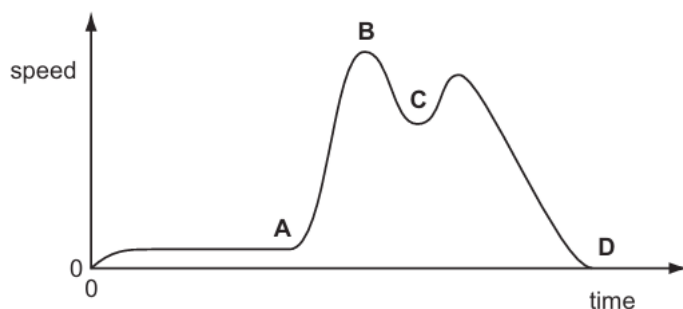
What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

43. M/J/12/12/Q3

A cyclist travels along a hilly road without using the pedals or brakes. Air resistance and friction are negligible. The speed/time graph of the cyclist is shown.

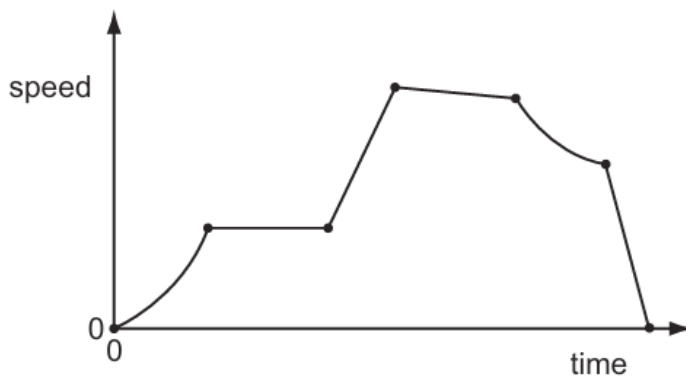
At which point did he reach the bottom of the first hill?



44. M/J/12/11/Q3

The speed-time graph represents the journey of a car.

The dots separate different sections of the journey. There are six different sections.



How many sections represent the car moving with non-uniform acceleration?

A 0

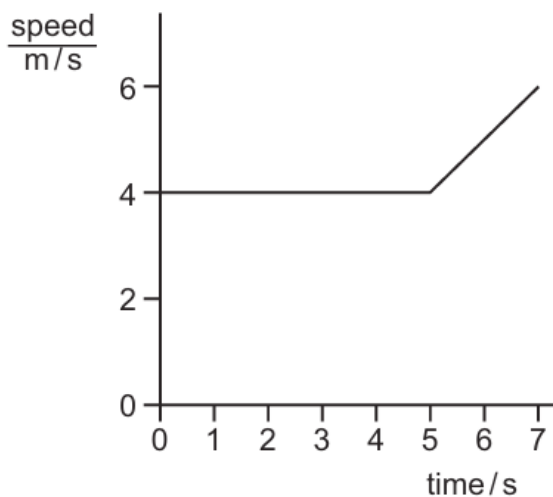
B 1

C 2

D 3

45. O/N/11/12/Q3

The graph shows part of a journey made by a cyclist.



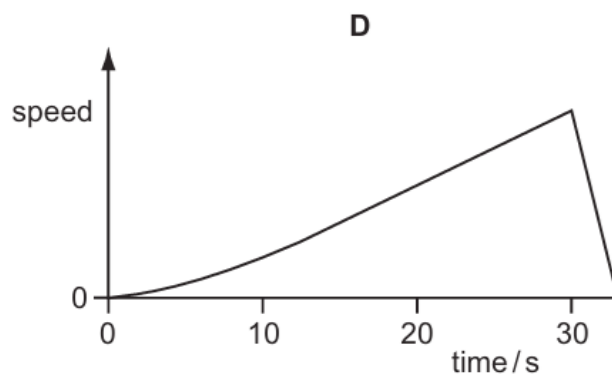
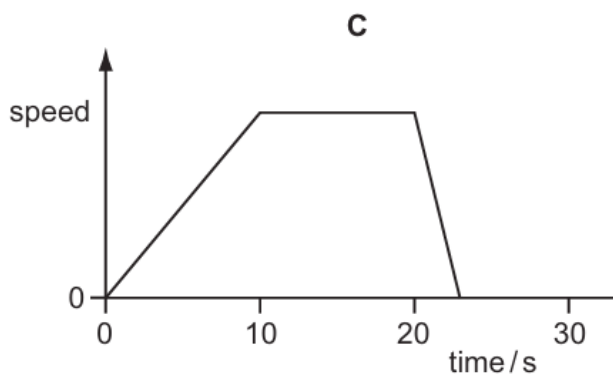
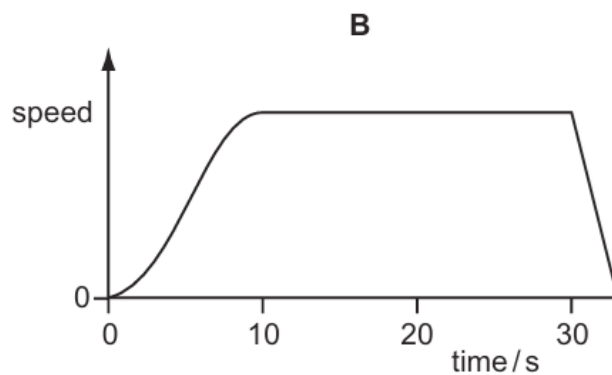
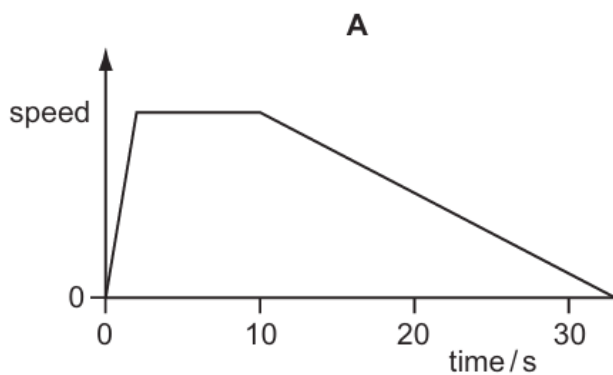
How far did the cyclist travel in 7 s?

- A** 28 m **B** 30 m **C** 32 m **D** 42 m

46. M/J/11/12/Q5

A car accelerates from traffic lights for 10 s. It stays at a steady speed for 20 s and then brakes to a stop in 3 s.

Which graph shows the journey?



47. M/J/11/12/Q4

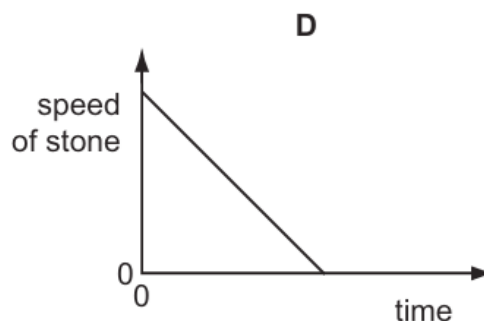
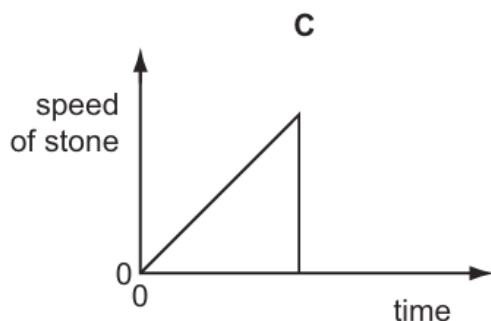
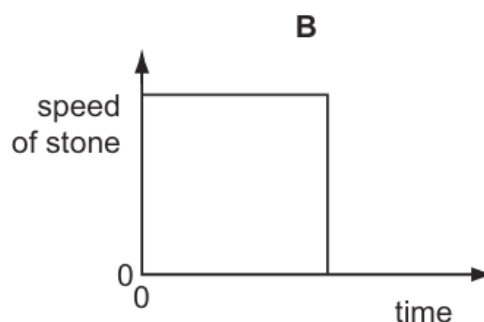
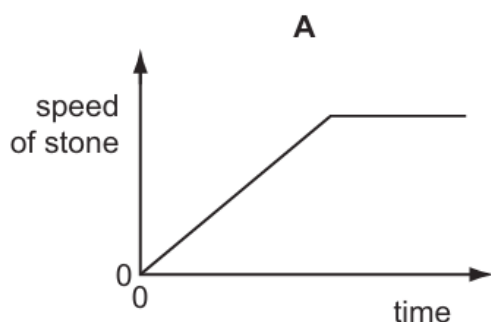
Which vehicle has an acceleration of 5 m/s^2 ?

- A** a bicycle, when its speed changes from rest to 2.5 m/s in 2 s
- B** a car, when its speed changes from rest to 15 m/s in 5 s
- C** a lorry, when its speed changes from rest to 20 m/s in 15 s
- D** a motorbike, when its speed changes from rest to 50 m/s in 10 s

48. O/N/10/12/Q2

A small stone is dropped from the top of a ladder, falls and hits the ground. It does not rebound.

Which speed-time graph is correct?



49. M/J/10/12/Q4

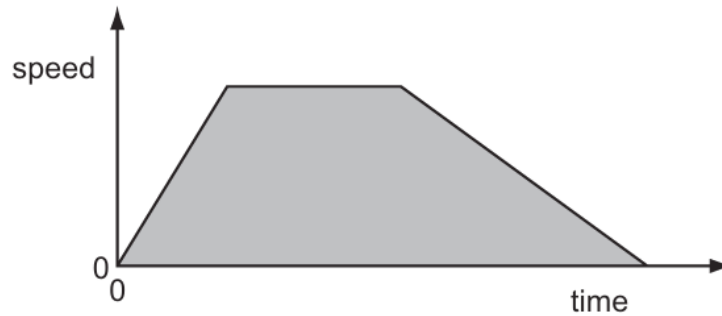
A student measures the speed of a trolley. At one instant, the speed of the trolley is 1.0 m/s and two seconds later the speed is 4.0 m/s .

What is the acceleration of the trolley?

- A** 1.5 m/s^2 **B** 2.0 m/s^2 **C** 2.5 m/s^2 **D** 5.0 m/s^2

50. M/J/10/12/Q5

The speed-time graph shows the movement of a car.



What does the shaded area of the graph represent?

- | | |
|--|--|
| A the average acceleration of the car | C the total distance travelled by the car |
| B the average speed of the car | D the total travelling time of the car |

51. M/J/10/12/Q4

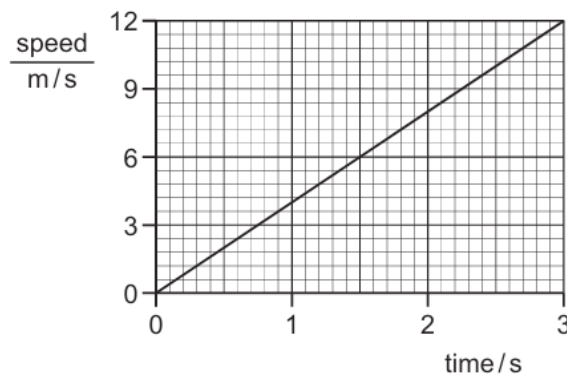
A student measures the speed of a trolley. At one instant, the speed of the trolley is 1.0 m/s and two seconds later the speed is 4.0 m/s .

What is the acceleration of the trolley?

- A** 1.5 m/s^2 **B** 2.0 m/s^2 **C** 2.5 m/s^2 **D** 5.0 m/s^2

52. O/N/09/Q4

The graph shows the speed of a car as it moves from rest.

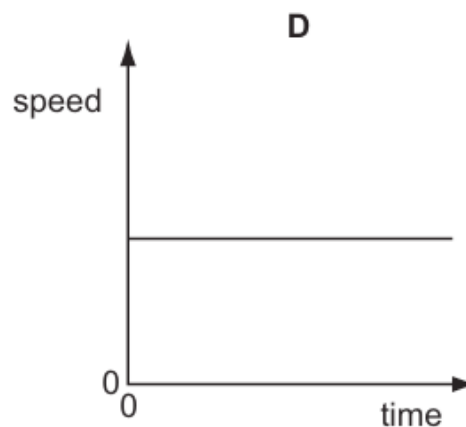
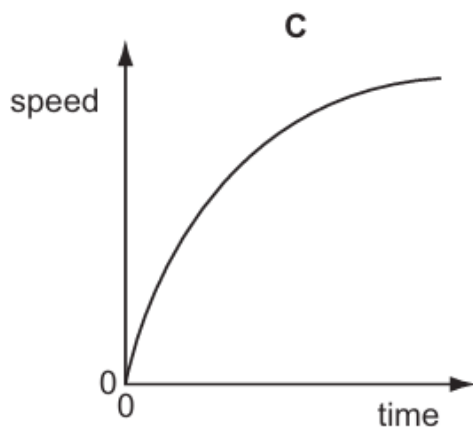
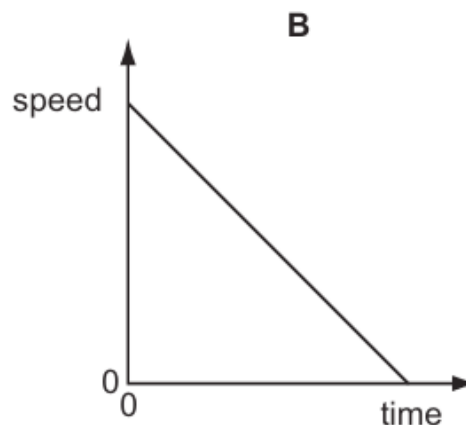
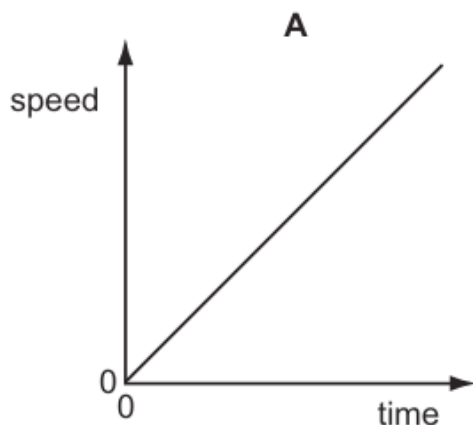


What is the average speed of the car during the first 3 s?

- A** 4 m/s **B** 6 m/s **C** 18 m/s **D** 36 m/s

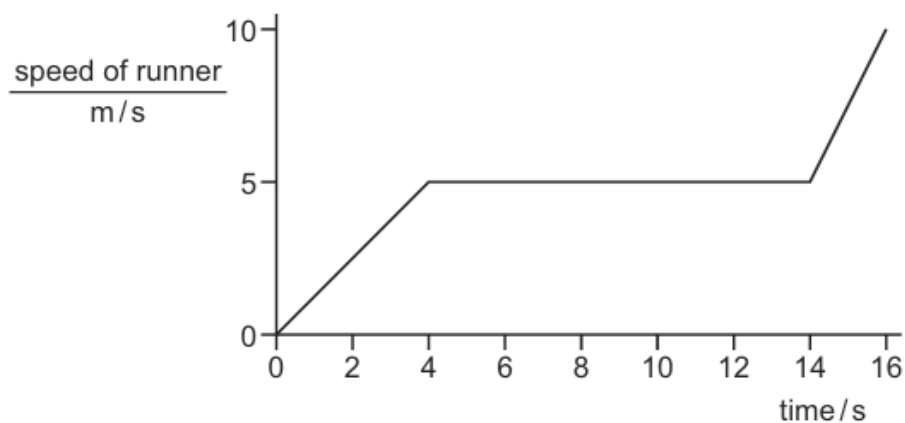
53. O/N/09/Q3

Which speed-time graph shows an object moving with non-uniform acceleration?



54. M/J/09/Q4

The graph shows the speed of a runner during a race.



What is the distance travelled by the runner during the race?

- A** 50 m **B** 65 m **C** 75 m **D** 90 m

55. O/N/08/Q3

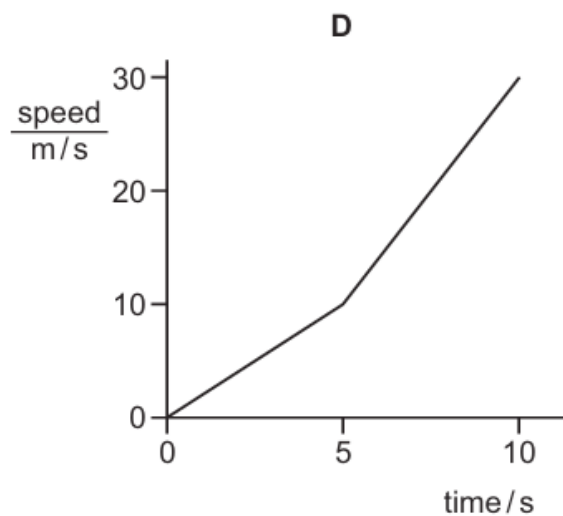
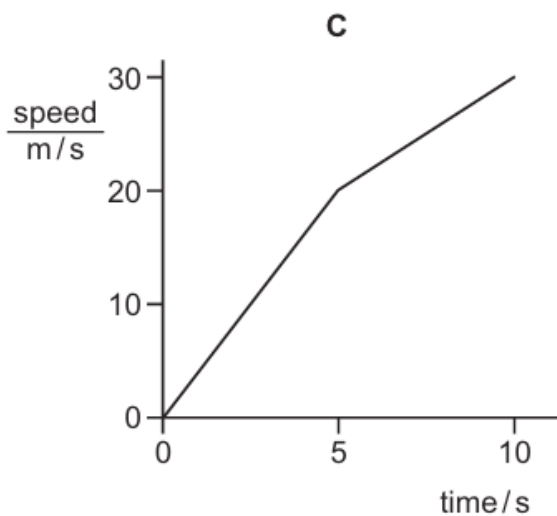
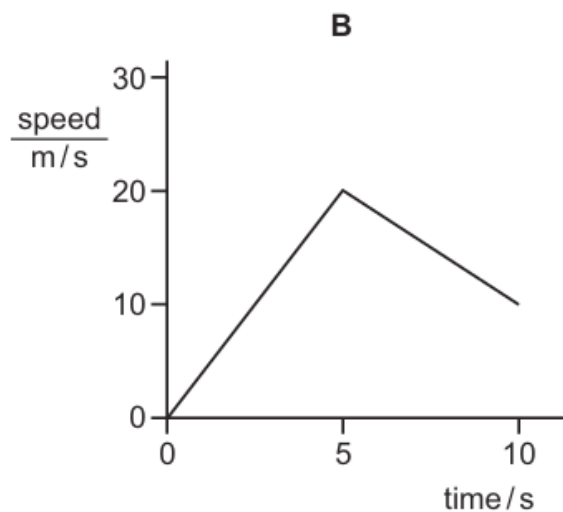
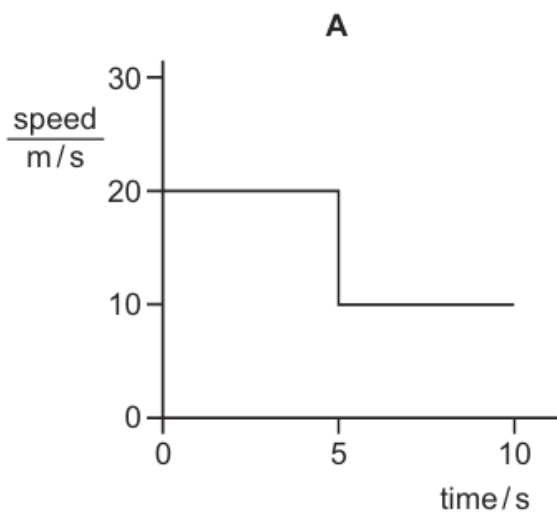
How is the motion of a body affected by balanced and unbalanced forces acting on it?

	balanced forces	unbalanced forces
A	velocity changes	velocity changes
B	velocity changes	velocity constant
C	velocity constant	velocity changes
D	velocity constant	velocity constant

56. M/J/08/Q4

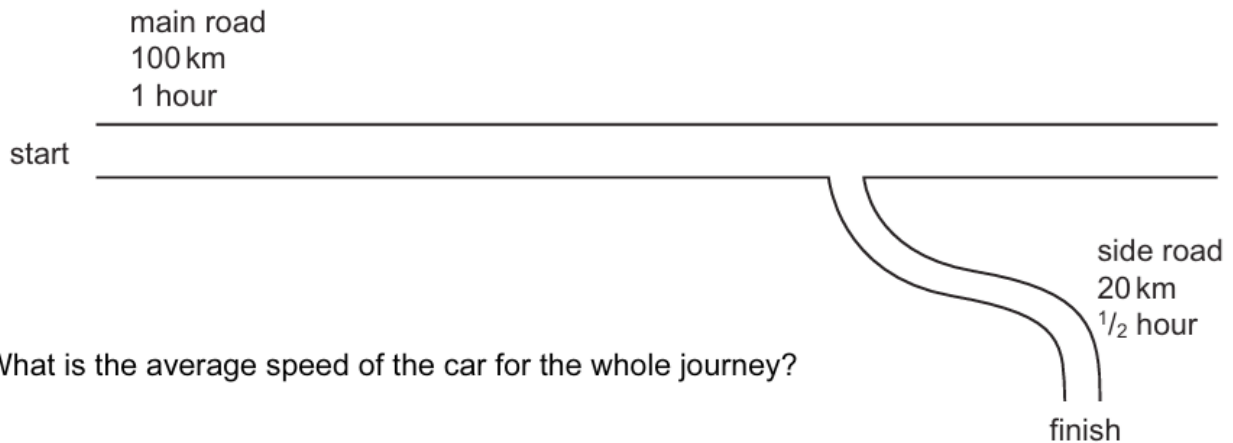
A skier is travelling downhill. The acceleration on hard snow is 4 m/s^2 and on soft snow is 2 m/s^2

Which graph shows the motion of the skier when moving from hard snow to soft snow?



57. O/N/08/Q2

A car takes 1 hour to travel 100 km along a main road and then $\frac{1}{2}$ hour to travel 20 km along a side road.

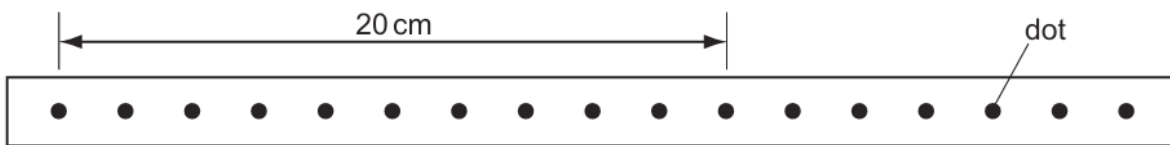


What is the average speed of the car for the whole journey?

- A** 60 km/h **C** 80 km/h
B 70 km/h **D** 100 km/h

58. O/N/06/Q5

The diagram shows a strip of paper tape that has been pulled under a vibrating arm by an object moving at constant speed. The arm was vibrating regularly, making 50 dots per second.

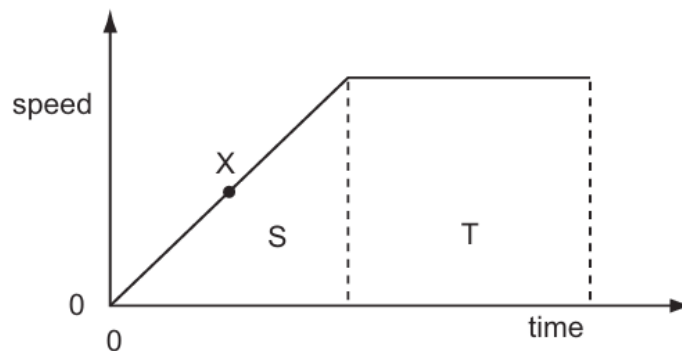


What was the speed of the object?

- A** 2.0 cm/s **B** 5.0 cm/s **C** 100 cm/s **D** 200 cm/s

59. O/N/05/Q4

The graph illustrates the motion of an object.



Which feature of the graph represents the distance travelled by the object whilst moving at a constant speed?

- A** area S **C** area T
B area S + area T **D** the gradient at point X

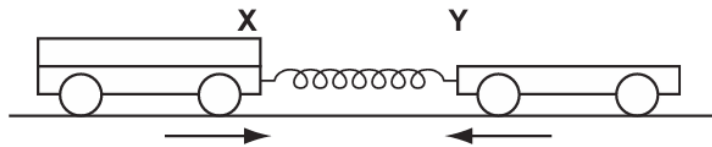
60. O/N/05/Q3

What **must** change when a body is accelerating?

- | | |
|---------------------------------------|-----------------------------------|
| A the force acting on the body | C the speed of the body |
| B the mass of the body | D the velocity of the body |

61. O/N/04/Q4

Trolley **X** and trolley **Y** are joined by a stretched spring. Trolley **X** has twice the mass of trolley **Y**.



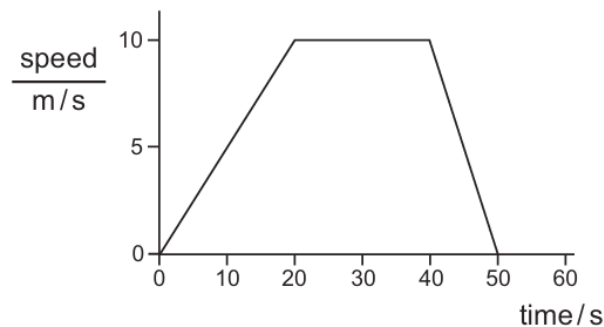
When the trolleys are released, the acceleration of **X** is 2 m/s^2 to the right.

What is the initial acceleration of trolley **Y** to the left?

- | | | | |
|----------------------------|----------------------------|----------------------------|----------------------------|
| A 1 m/s^2 | B 2 m/s^2 | C 3 m/s^2 | D 4 m/s^2 |
|----------------------------|----------------------------|----------------------------|----------------------------|

62. M/J/04/Q3

The graph shows the movement of a car over a period of 50 s.



What was the distance travelled by the car while its speed was increasing?

- | | | | |
|---------------|---------------|----------------|----------------|
| A 10 m | B 20 m | C 100 m | D 200 m |
|---------------|---------------|----------------|----------------|

63. M/J/04/Q2

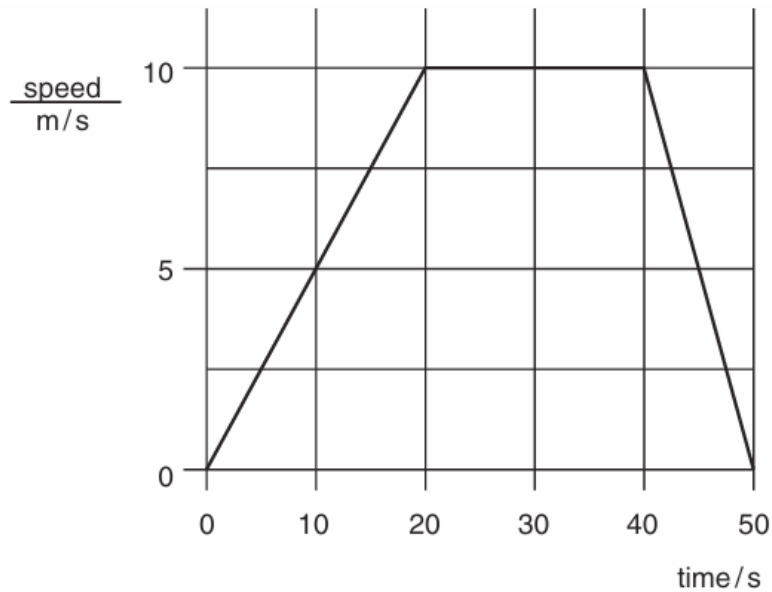
Which quantity **X** is calculated using this equation?

$$X = \frac{\text{change in velocity}}{\text{time taken}}$$

- | | |
|---------------------------|-----------------------------|
| A acceleration | C distance travelled |
| B average velocity | D speed |

64. O/N/03/Q3

The graph shows the movement of a car over a period of 50 s.



What was the distance travelled by the car during the time when it was moving at a steady speed?

- A 10 m C 200 m
B 100 m D 400 m

65. O/N/03/Q2

Which of the following **defines** acceleration?

- A $\frac{\text{change in velocity}}{\text{time taken}}$ C $\frac{\text{change in distance}}{\text{time taken}}$
B $\frac{\text{change in speed}}{\text{time taken}}$ D $\frac{\text{change in distance in a fixed direction}}{\text{time taken}}$

66. O/N/02/Q3

A car driver takes a total of two hours to make a journey of 75 km. She has a coffee break of half an hour and spends a quarter of an hour stationary in a traffic jam.

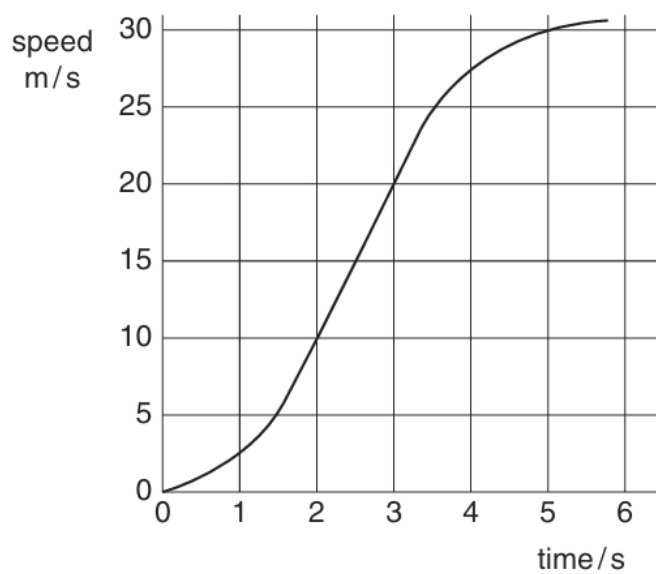
At what average speed must she travel during the rest of the time if she wants to complete the journey in the two hours?

- A 38 km/h B 50 km/h C 60 km/h D 75 km/h

67. M/J/02/Q3

The graph shows the speed of a car as it accelerates from rest.

During part of this time the acceleration is uniform.



What is the size of this uniform acceleration?

- A** 5 m/s^2 **B** 6 m/s^2 **C** 10 m/s^2 **D** 20 m/s^2